

KHOLMSK, Russian Federation

WMO: 321280

Lat: 47.050N

Lon: 142.050E

Elev: 44

StdP: 100.80

Time Zone: 11.00 (E11)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.2	-13.6	-20.3	0.6	-13.6	-18.6	0.7	-12.7	12.3	-7.6	11.3	-6.8	2.5	90	0.598

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.5	23.4	20.3	22.4	19.6	21.5	18.8	21.1	22.8	20.2	21.8	19.4	21.0	3.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.5	15.2	22.2	19.6	14.4	21.4	18.7	13.6	20.6	61.8	22.9	58.6	22.0	55.8	21.0	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
10.6	9.1	8.0	DB	-16.4	25.4	2.1	1.3	-17.8	26.4	-19.0	27.1	-20.2	27.9	-21.7	28.8	(4)
			WB	-17.1	21.8	1.9	1.1	-18.5	22.6	-19.5	23.3	-20.6	23.9	-21.9	24.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.3	-7.7	-7.1	-2.6	3.3	8.2	12.5	16.7	18.7	15.7	9.0	1.1	-5.2
	DBStd	9.73	3.55	3.74	3.45	3.11	3.22	2.76	2.52	2.36	2.77	3.41	4.28	3.55
	HDD10.0	2502	548	478	390	203	75	8	0	0	1	59	269	471
	HDD18.3	4816	806	712	648	452	314	176	61	24	86	289	518	729
	CDD10.0	784	0	0	0	1	19	84	209	270	171	29	1	0
	CDD18.3	56	0	0	0	0	0	1	12	36	7	0	0	0
	CDH23.3	33	0	0	0	0	0	1	6	24	2	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	3.3	3.6	3.6	3.8	3.5	3.1	2.3	2.1	2.6	3.1	3.9	4.2	3.9
Precipitation	PrecAvg	795	54	36	29	46	65	53	91	98	99	82	72	70
	PrecMax	1070	108	76	79	140	123	139	252	344	196	205	207	150
	PrecMin	578	18	4	1	9	28	2	9	21	24	24	14	17
	PrecStd	141	22	19	19	27	26	29	48	68	42	39	35	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.9	3.6	6.7	14.1	18.8	21.5	24.2	25.2	23.3	18.1	13.0	5.8
		MCWB	0.5	2.2	3.8	9.4	12.6	16.1	19.7	21.0	20.1	14.7	11.2	4.6
	2%	DB	-0.1	1.7	4.6	11.4	16.6	19.0	22.4	23.9	21.8	16.5	10.7	3.3
		MCWB	-1.6	0.3	2.4	7.1	11.5	14.8	19.1	20.9	18.6	13.5	9.0	1.8
	5%	DB	-1.5	0.1	3.5	9.5	14.6	17.8	21.4	23.1	20.9	15.2	8.9	1.5
		MCWB	-2.8	-1.4	1.7	6.0	10.4	14.4	18.6	20.4	17.8	12.5	6.8	0.0
	10%	DB	-2.6	-1.6	2.3	7.9	13.2	16.7	20.4	22.2	19.8	14.0	7.3	-0.1
		MCWB	-3.9	-3.0	0.6	5.0	9.7	13.9	18.1	19.7	16.9	11.4	5.3	-1.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.7	2.4	4.4	10.5	14.1	17.7	21.2	22.6	20.8	15.5	11.2	4.5
		MCDB	1.8	3.2	5.9	13.2	16.9	19.7	22.8	23.9	22.6	17.5	12.6	5.7
	2%	WB	-1.4	0.5	2.9	7.9	12.2	16.2	20.0	21.7	19.4	14.0	9.0	2.0
		MCDB	-0.3	1.6	4.1	10.6	15.2	18.0	21.4	23.1	21.4	16.0	10.4	3.1
	5%	WB	-2.7	-1.3	1.8	6.4	11.2	15.2	19.3	21.0	18.3	12.9	7.0	0.2
		MCDB	-1.6	0.0	3.3	8.9	13.8	17.0	20.7	22.6	20.3	14.8	8.5	1.3
	10%	WB	-3.9	-2.9	0.8	5.2	10.2	14.3	18.6	20.2	17.3	11.7	5.3	-1.4
		MCDB	-2.7	-1.6	2.3	7.5	12.6	16.0	20.0	21.7	19.5	13.7	7.3	-0.2
Mean Daily Temperature Range	5% DB	MDBR	3.8	4.3	4.3	4.8	5.3	4.8	4.4	4.5	5.1	5.0	4.3	3.8
		MCDBR	4.7	5.4	5.1	6.7	7.2	6.4	5.2	4.8	5.6	5.6	5.6	5.7
		MCWBR	4.5	5.2	4.6	4.7	4.4	4.1	3.4	3.5	4.3	5.1	5.3	5.7
	5% WB	MCDBR	4.7	5.4	5.0	6.3	6.1	5.0	4.4	4.0	5.0	5.2	5.8	5.8
		MCWBR	4.6	5.4	4.8	4.8	4.3	3.8	3.4	3.4	4.5	5.3	6.0	6.0
Clear-Sky Solar Irradiance	taub	0.322	0.365	0.419	0.497	0.523	0.500	0.464	0.440	0.397	0.404	0.378	0.341	
	taud	2.078	1.958	1.891	1.831	1.873	2.033	2.212	2.289	2.339	2.145	2.097	2.043	
	Ebn at Noon	720	761	779	756	754	777	802	806	809	718	644	634	
	Edh at Noon	94	132	166	194	193	166	137	122	106	110	94	88	
All-Sky Solar Radiation	RadAvg	1.19	2.08	3.38	4.52	4.84	4.77	4.46	4.23	3.82	2.52	1.38	0.93	
	RadStd	0.14	0.20	0.29	0.48	0.45	0.53	0.54	0.38	0.36	0.17	0.13	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page